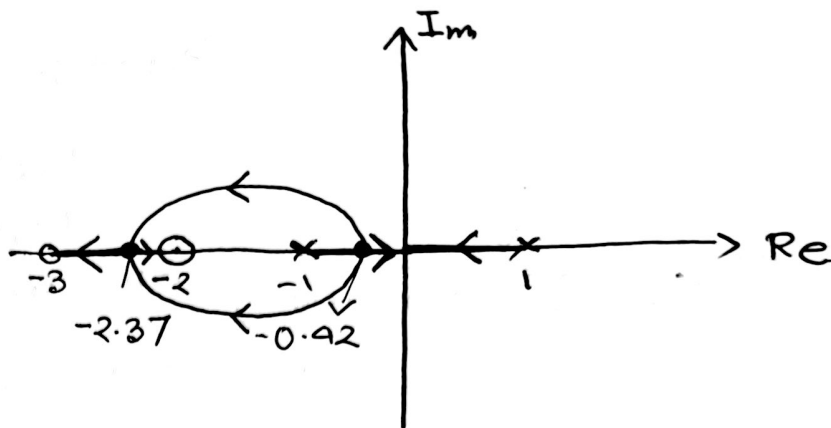


$$3. \quad G(s) = \frac{k(s+2)(s+3)}{(s+1)(s-1)}$$

$$1. \quad z_1 = -2, \quad z_2 = -3;$$

$$p_1 = -1, \quad p_2 = 1$$

$$\text{Total loci} = \max(P, Z) = 2$$



3. BAP

$$\text{ch. eqn } 1 + G(s)H(s) = 0$$

$$\Rightarrow 1 + \frac{k(s+2)(s+3)}{(s+1)(s-1)} \times 1 = 0$$

$$\Rightarrow s^2 - 1 + k(s^2 + 5s + 6) = 0$$

$$\Rightarrow k = -\frac{s^2 - 1}{s^2 + 5s + 6}$$

$$\frac{d(A/B)}{ds} = \frac{B \frac{dA}{ds} - A \frac{dB}{ds}}{B^2}$$

$$\frac{dk}{ds} = 0 \Rightarrow \frac{(s^2 + 5s + 6)(2s) - (s^2 - 1)(2s + 5)}{(s^2 + 5s + 6)^2} = 0$$

$$\Rightarrow 2s^3 + 10s^2 + 12s - 2s^3 - 5s^2 + 2s + 5 = 0$$

$$\Rightarrow 5s^2 + 14s + 5 = 0$$

$$\Rightarrow s = \frac{-14 \pm \sqrt{14^2 - 4ac}}{2 \times 5}$$

$$= -0.42, -2.37$$